Libspace backend system
Ideation Phase
Define the Problem Statements

Date	8 March 2026
Team ID	69aa7af9d483472145093618
Project Name	Libspace
Maximum Marks	2 Marks

Customer Problem Statement – Libspace Backend System

The Customer Problem Statement helps identify the challenges faced by users while managing books in a digital system. Understanding the user's perspective allows the team to design a backend application that is secure, structured, and efficient.

Many small libraries, students, and individual users struggle to manage book records manually or through unorganized digital methods. Existing systems are often:

- Overly complex
- Paid or subscription-based
- Not secure
- Lacking proper user-specific data access
- Difficult to maintain

There is a need for a simple and secure backend system that allows users to manage books efficiently while ensuring data privacy and authentication control.

Identified User Challenges

1. Lack of secure authentication mechanisms
2. No centralized system to store and manage books
3. Inability to restrict access to user-specific data
4. Difficulty in updating and deleting records
5. Risk of data inconsistency or duplication

Customer Needs

Users require a system that:

- Allows secure user registration and login
- Ensures passwords are encrypted and protected
- Restricts access to authorized users only
- Provides full CRUD operations for books
- Maintains proper database structure
- Ensures reliable error handling

Project Focus Based on Customer Needs

To address these challenges, the team decided to build **Libspace – Backend Book Management System**, which:

- Implements JWT-based authentication
- Provides secure login and logout functionality
- Enables users to perform CRUD operations on books
- Ensures user-specific data isolation
- Follows RESTful API standards
- Implements middleware for route protection

Outcome of the Customer Problem Analysis

By analyzing customer pain points, the project focuses on delivering:

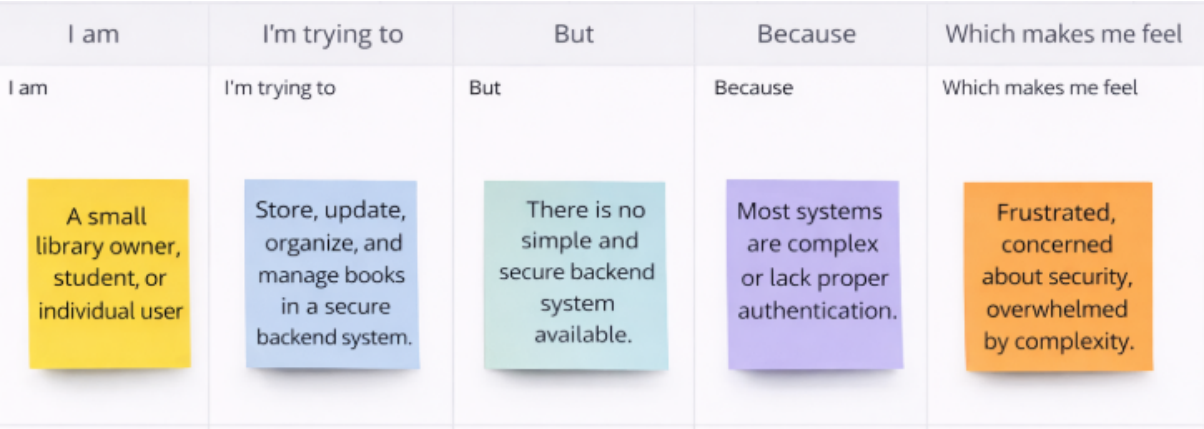
- Secure backend architecture
- Structured database management

- Clean API design
- Scalable system foundation
- Real-world backend implementation practices

The ideation phase ensured that the system is not just technically functional but also aligned with real user problems and practical backend development goals.

I am	Describe customer with 3-4 key characteristics - *who are they?*	Describe the customer and their attributes here
I'm trying to	*List their outcome or "job" the care about - what are they trying to achieve?*	List the thing they are trying to achieve here
but	Describe what problems or barriers stand in the way – *what bothers them most?*	Describe the problems or barriers that get in the way here
because	Enter the "root cause" of why the problem or barrier exists – *what needs to be solved?*	Describe the reason the problems or barriers exist
which makes me feel	Describe the emotions from the customer's point of view – *how does it impact them emotionally?*	Describe the emotions the result from experiencing the problems or barriers

Reference: https://miro.com/templates/customer-problem-statement/



Problem Statement	I am (Customer)	I'm trying to	But	Because	Which makes me feel

PS-1	A small library owner	Manage and organize book records digitally	Existing tools are complex or require paid subscriptions	I need a simple backend system to securely add, update, and delete books	Frustrated managing records manually
PS-2	A student managing personal books	Keep track of books in one centralized system	There is no secure system that restricts access to my data	I want a login-based system where only I can manage my books	Concerned about data privacy
PS-3	A individual user	Update and delete book information easily	Many systems do not provide proper authenticatio n or data isolation	I need a secure backend with JWT authentication and protected routes	Anxious about unauthorized access
PS-4	A beginner user with limited technical knowledge	Use a simple system to manage books without complexity	Most available platforms focus more on UI and less on backend security	I want a lightweight and secure system that is easy to use	Overwhelmed by complicated software